

Presentation 3

What did each of the team members do?

Henry: Coder optional task 2, improved data leakage detection, made 5 non-overlapping premade splits to train on, split on original case id not apparent id. Evaluated models on those 5 splits.

Vicky: Remove dropout from the attention MLP; Use gated attention to enhance non-linear feature capture.

Zijin: Visualized the performance of the best model, including training and validation loss curves, prediction deviation plots, and effective patches of misclassified cases. Understand the patching code from the previous quarter, and identify some potential issues.

Abhi:

How does it help the project?

Henry: Better splitting logic for our problem. Generalizable script for making data splits.

Vicky: Our implementation of Gated Attention and Entropy Regularization focused on reducing false-positive triggers in benign cases. We achieved a 71.4% Benign Recall on the test set.

Zijin: It provides a framework for visualizing model performance across different models, helping identify potential issues during training. The exploration of the patching code also enables visualization of high-attention regions in histopathology images, improving model interpretability.

Abhi:

Issues faced that haven't been resolved:

- Loss graph doesn't work if resuming from checkpoint
- Actually random splits underperform in getting representative high-grade ratios in all sets
- Visualize the high attention patches: missing .png masks, unmatched names

Proposed next steps:

- Continue working on the patching code to resolve file structure and naming inconsistencies

References

Henry: Team4/Krish's base code, Team5's case splits, AI tools

Vicky: Team4/Krish's base code, AI tools

Zijin: AI tools,

https://github.com/arvindkrishna87/STAT390_CMIL_Fall2025/tree/main/Code/Patching%20code

Links to the presentation & code:

- **Presentation:** [Presentation3 Team6.pdf](#)
- **Code:** [Code3 NewBaseCode 24Feb Team6](#)
(Henry & Zijin changes)
- **Code:** [Code3 NewBaseCode 3Mar Team6](#)
(Vicky & Zijin changes)